INCH - POUND

MIL-PRF-26542/1E
16 July 2002
SUPERSEDING
MIL-PRF-26542/1D
30 May 1997

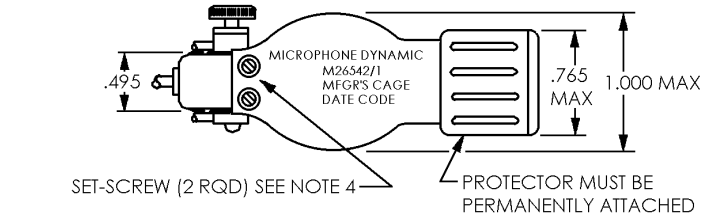
PERFORMANCE SPECIFICATION SHEET

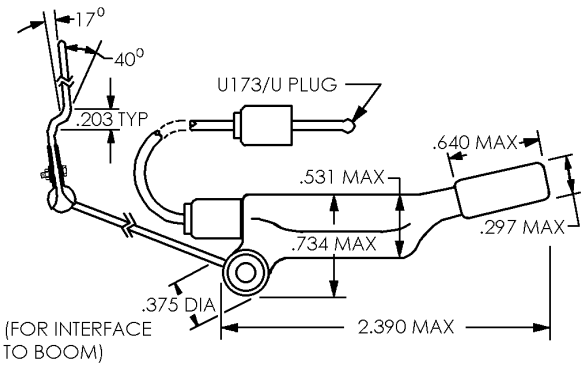
* MICROPHONE AND MICROPHONE ASSEMBLIES,
M26542/1, M26542/1-01

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.

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Inches	mm
.203	5.16
.297	7.54
.375	9.53
.495	12.57
.531	13.49
.640	16.26
.734	18.64
.765	19.43
1.000	25.40

NOTES:

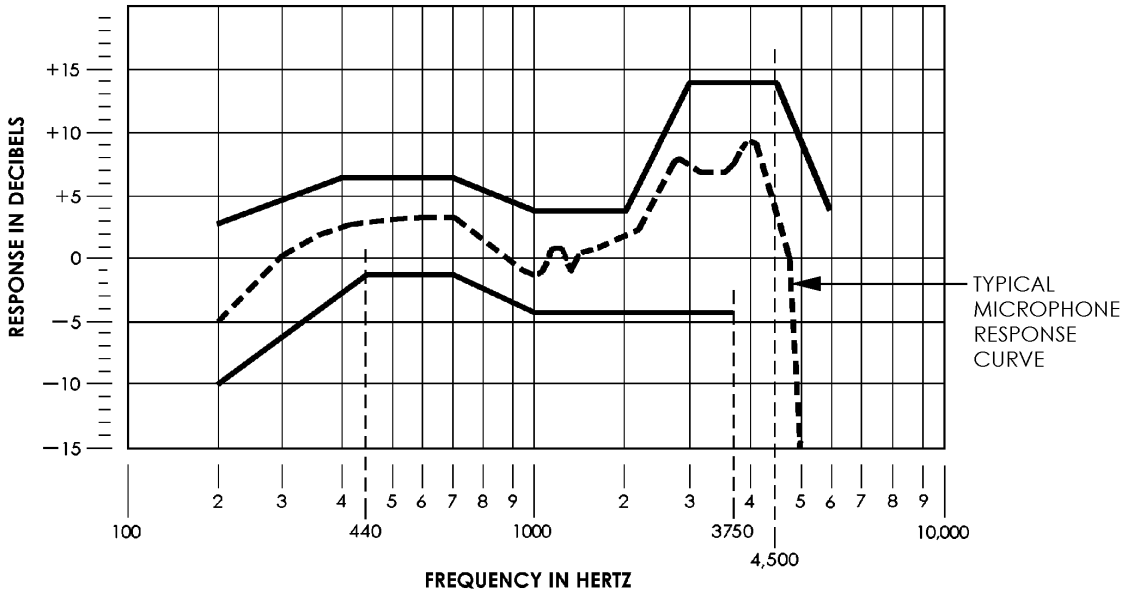
1. Quantity and configuration of sound ports optional.
2. * The microphone element shall be marked with the same Part or Identifying Number (PIN) (i.e., M26542/1). The combined microphone-cable assembly PIN (i.e., M26542/1-01) shall appear on the packaging for that assembly in accordance with MIL-STD-129. Placement on surface shown is optional.
3. Dimensions are in inches. Tolerance is ± .015 inch, unless otherwise specified.
4. Screws shall hold the element securely, shall be either slotted or allen type, and shall not protrude above the surface of the element.
5. Angular requirements of boom shall be met to provide interface to headset, and for adjustability.

FIGURE 1. Microphone assembly.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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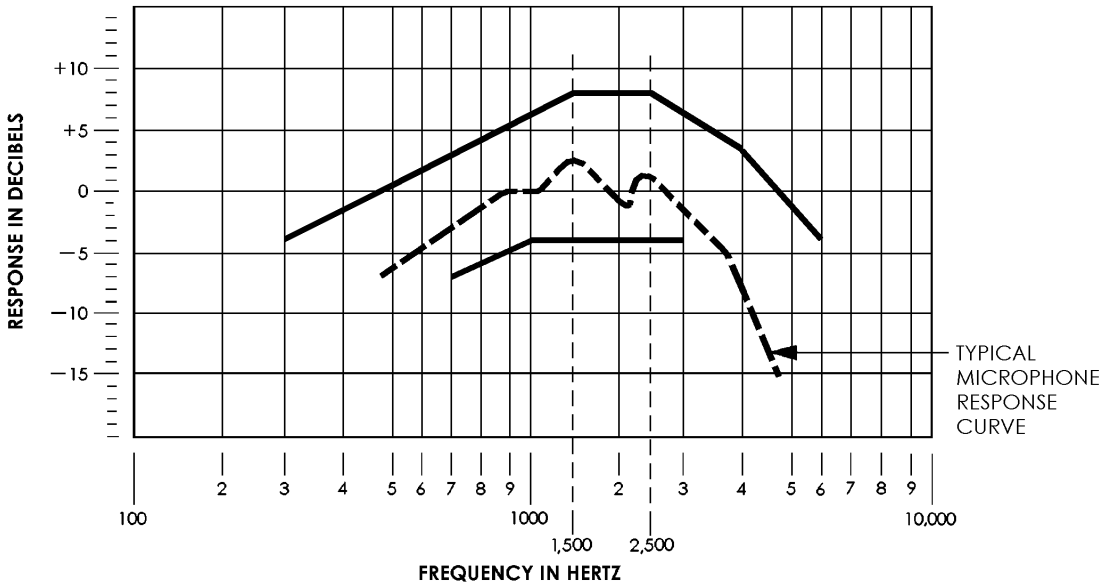
Frequency points	200 Hz	400 Hz	700 Hz	1,000 Hz	3,000 Hz	3,750 Hz	4,400 Hz	6,000 Hz
Upper Limits (dB)	+2.50	+7.0	+7.0	+3.75	+13.75	+13.75	+13.75	+3.0
Lower Limits (dB)	-10.0	-2.60 1/	-1.50	-4.25	-4.25	-4.25	---	---

1/ dB limits between key break point are calculated, using slope method.

FIGURE 2. Frequency response at ground level.

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Frequency points	300 Hz	700 Hz	1,000 Hz	1,500 Hz	2,500 Hz	3,000 Hz	4,000 Hz	6,000 Hz
Upper Limits (dB)	-4.0	+2.32 (See note)	+5.62 (See note)	+8.0	+8.0	+6.26 (See note)	+3.50	-4.0
Lower Limits (dB)	---	-7.0	-4.0	-4.0	-4.0	-4.0	---	---

NOTE: dB limits between key break point are calculated, using slope method.

FIGURE 3. Frequency response at 25,000 feet.

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REQUIREMENTS

Component parts.

Boom:

Dimensions: Dimensions of the boom shall be in accordance with USAF Drawing 67B1854, for interchangeability with the next-higher-assembly headset-microphone.

Color: Boom shall be the same color as the microphone element.

* Finish: Shall be in accordance with MIL-PRF-26542, (see boom finish requirements).

* Operation: Shall be in accordance with MIL-PRF-26542, (see boom-operating force requirements).

Material: Shall be constructed from a high-strength, corrosion-resistant metal, meeting or exceeding the environmental and durability requirements of MIL-PRF-26542.

* Cable assembly: The cable assembly shall be PIN M22442/36-1 in accordance with either requirement 1/ for interface to oxygen gear, interchangeability, and environmental performance. The microphone element shall provide a complete electrical and mechanical interface with the cable assembly.

Plug assembly: U-173/U, in accordance with USAF Drawing 57B12662, or electrically and mechanically compatible part.

Weight: 45 grams, maximum.

Performance:

Sensitivity at ground level: 49.77 dB - 55.79 dB (re 1 µV) or 307.96 µV - 615.88 µV with a Sound Pressure Level (SPL) input of 28 dynes/cm² at 1 kHz, when tested with the microphone sound port .187 ± .015 inch (4.75 ± .38 mm) from, and coaxial with, the opening of the artificial voice.

Sensitivity at altitude: Shall be within ±3 dB of initial ground level sensitivity tested at a simulated 25,000 feet.

* Frequency response envelope: Shall be as shown on figures 2 and 3, when tested with the microphone sound port .187 ± .015 inch (4.75 ± .38 mm) from, and coaxial with, the opening of the artificial voice. The response curves generated shall be on the same scale as shown on figures 2 & 3. The response curve shall not exceed the upper and lower limit curves of the stationary Frequency Response Envelope, within the frequency ranges identified in the appropriate chart (see figures 2 and 3).

Impedance: 150.0 ohms ±1.5 ohms.

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1/ The manufacturer shall meet the requirements of MIL-DTL-22442 in accordance with options specified in MIL-PRF-26542 for ‘cable assemblies’.

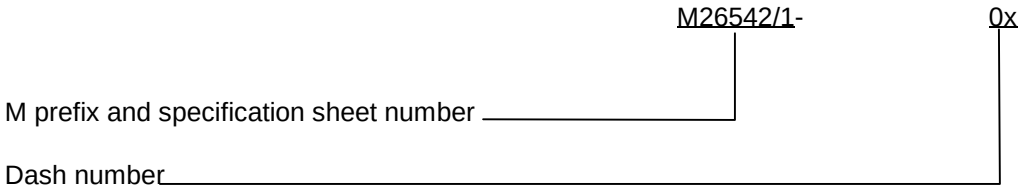
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Total harmonic distortion: Between 200 Hz to 6,000 Hz, 5 percent at 105 dB relative to 20 micropascals.

Resistive load: 150 ohms ±1.5 ohms.

Intended use: The microphone, including the boom and cord assembly, is intended for use in headset-microphones requiring noise-canceling communications at both ground level and altitude.

PIN: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code, as outlined in table I.



The microphone assembly shall be tested in accordance with the tests listed in table II.

* TABLE I. PIN designations.

PIN	Characteristics
M26542/1	Supplied with microphone element only.
M26542/1-01	Supplied with microphone element, boom, and cable assembly specified.

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* TABLE II. Parameter applicability.

Inspection	Qualification	Group "A"	Group "B"	Group "C"
Group I				
Visual and mechanical inspection	X	X		
Sensitivity at ground level	X	X		
Sensitivity at altitude	X			
Frequency response at ground level	X	X		
Frequency response at altitude	X			
Impedance	X	X		
Noise cancellation characteristic	N/A			
Effect of external magnetic field	X			
Stray magnetic field	X			
Linearity	X			
Talk-out	X	X		
Dielectric withstanding voltage	X			
Signal-to-noise	X		X	
Distortion	X		X	
Interchangeability	X		X	
Group II				
Thermal shock	X			X
Humidity	X			X
Drop	X			X
Pressure equalization	X			X
Explosive decompression	X			X
Salt fog	X			X
Group III				
Vibration	X			X
Bounce	X			X
Altitude	X			X
Moisture barrier seal	X			X
Immersion	N/A			
Group IV				
Fungus	X			
Group V				
Gun blast	N/A			
Boom finish	X		X	
Boom operating force	X		X	

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force -11
DLA - CC

Review activities:
Army - AR, AT, AV, CR4
Navy - AS, OS
Air Force - 99

Preparing activity:
DLA - CC

(Project 5965-0351-001)

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